

DECA Section 2

2025年11月11日 星期二

08:42

DECA Ss.

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Lab. Charles Wu.

Reference: DECA section 2.

Aim: Building and learning a half-adder; full adder; ripple carry adder; and a subtractor.

Procedure

- Create a half-adder.

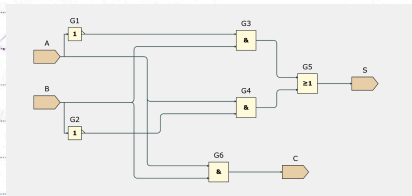
Half-adder adds 2 one-bit digits and produce a sum and a carry-out.

$$S = A\bar{B} + \bar{A}B = A \oplus B.$$

$$C = AB.$$

Expect:

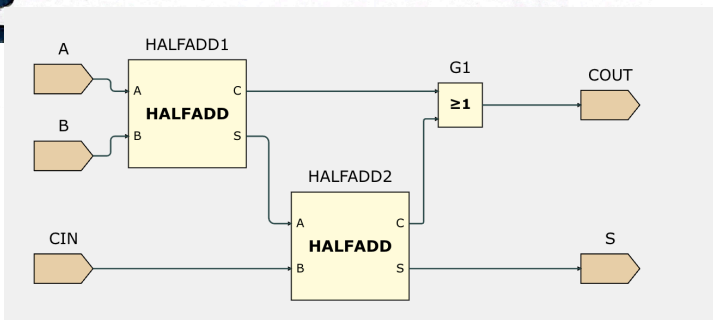
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



- Create a full-adder.

A full-adder works by combining A, B, and Cin to create a sum and a carry-out.

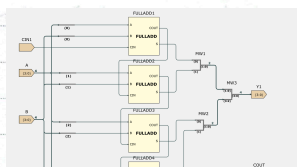
Cin can let full-adders to be chained together to become multi-bit.



- Ripple-carry adder.

Now we can do addition for multi-bits just by:

1. Select correct bus, and do a full-adder in each bit according
2. Connect Cin in higher bit to Cout in lower to pass the carry-out.
3. Merge and get final sum & carry-out.



• Subtraction

Subtraction can be done by adding the complement

e.g. $7-4 = 7+(19-4)+1 = 12 \rightarrow \text{Cout} = 1$ (Leave)
 $S = 2 \checkmark$

For binary: $A-B = A+(\bar{B}+1)$.

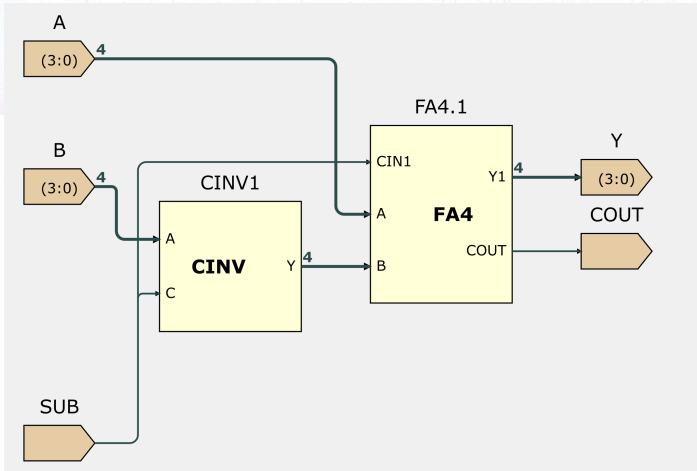
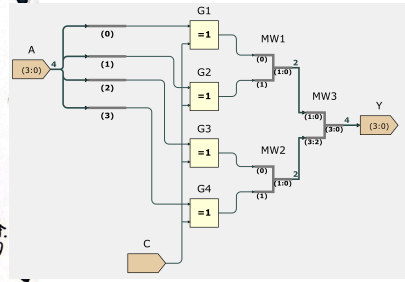
First, design an inverter "CINV" for B.

$\therefore$ For subtraction, Cin in "CINV" and "full-adder" both = 1.

For addition, Cin are both = 0.

$\therefore$ We can use a single SELECT to switch the adder/subtractor.

Note: For inverter: $\downarrow$ when Cin = 0, $B \oplus 0 = B$ $\leftarrow$
 $\downarrow$ when Cin = 1, $B \oplus 1 = \bar{B}$. $\leftarrow$



Do a random test, I test

	A	B	Sub	S	Cout
Adding	0101	0011	0	1000	0
	1100	1011	0	0111	1
Subtracting	1001	0100	1	0101	0
	0011	0111	1	1100	0

In subtracting Cout = 0 represents there's a borrow happened.
 Cout = 1, no.

• Challenge: Expand ALU with add, subtract, increment, and decrement.

which has: $Y = A+B$;

$A-B$;

$A+1$;

$A-1$;

In it, the first MUX select whether use A or B to do arithmetic operations with A by S0;

S1 controls add or sub;

S0 & S1 control which logic gate to use;

Finally, S2 determines whether to use logic gate or arithmetic which gives:

